Examples of Malware and CVEs

Foundations of Cybersecurity

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Persistence, Local Channels and Credentials

- This short appendix contain ten examples of
  - interplay in the use of CVEs and malware
  - ideas on how to use advanced schemes, like information hiding.
- The CVSS score and vector are also provided.

Exploit Delivery and Propagation

CVE-2010-2568 **Stuxnet propagation vector**	Used to deploy malware capable of stealth PLC manipulation and hidden process alteration in OT environments.	CVSS 9.3 AV:N/AC:L/PR:N/UI:R/ S:C/C:H/I:H/A:H
CVE-2014-4114 **Sandworm / BlackEnergy campaigns**	Exploited in ICS-oriented attacks relying on stealth command-and-control and protocol abuse.	CVSS 9.3 AV:N/AC:L/PR:N/UI:R/ S:C/C:H/I:H/A:H
CVE-2017-11882 **Microsoft Equation Editor exploit**	Widely used to deploy malware families retrieving payloads hidden via steganography from remote servers.	CVSS 7.8 AV:L/AC:L/PR:N/UI:R/ S:U/C:H/I:H/A:H
CVE-2021-40444 **MSHTML remote code execution**	Frequently used in staged malware delivery chains involving covert retrieval of payloads hidden inside legitimate content.	CVSS 8.8 AV:N/AC:L/PR:N/UI:R/ S:U/C:H/I:H/A:H

Covert Access and Command Exchange

CVE-2020-10148 **SolarWinds / SUNBURST**	Vulnerability used to deploy stealth malware leveraging DNS-based covert command-and-control and staged communications.	CVSS 9.8 AV:N/AC:L/PR:N/UI:N/ S:U/C:H/I:H/A:H
CVE-2021-26855 **ProxyLogon**	Exploited by APT groups for deploying stealth webshells and covert HTTP(S)-based exfiltration channels.	CVSS 9.8 AV:N/AC:L/PR:N/UI:N/ S:U/C:H/I:H/A:H
CVE-2021-44228 **Log4Shell**	Can trigger outbound DNS/LDAP callbacks enabling covert signaling, staged payload delivery, and stealth exfiltration.	CVSS 10.0 AV:N/AC:L/PR:N/UI:N/ S:C/C:H/I:H/A:H

Persistence, Local Channels and Credentials

CVE-2021-30747 **Apple Silicon “M1RACLES”**	Hardware covert channel exploiting undocumented Apple M1 registers for hidden inter-process communications.	CVSS 5.5 AV:L/AC:L/PR:L/UI:N/ S:C/C:L/I:L/A:N
CVE-2021-34527 **PrintNightmare**	Popular post-exploitation to deploy covert beacons and hidden persistence mechanisms.	CVSS 8.8 AV:N/AC:L/PR:L/UI:N/ S:U/C:H/I:H/A:H
CVE-2023-23397 **Outlook NTLM leak**	Enables stealth credential exfiltration through crafted reminders and hidden remote resource access.	CVSS 9.8 AV:N/AC:L/PR:N/UI:N/ S:U/C:H/I:H/A:H